

# Information Visualization

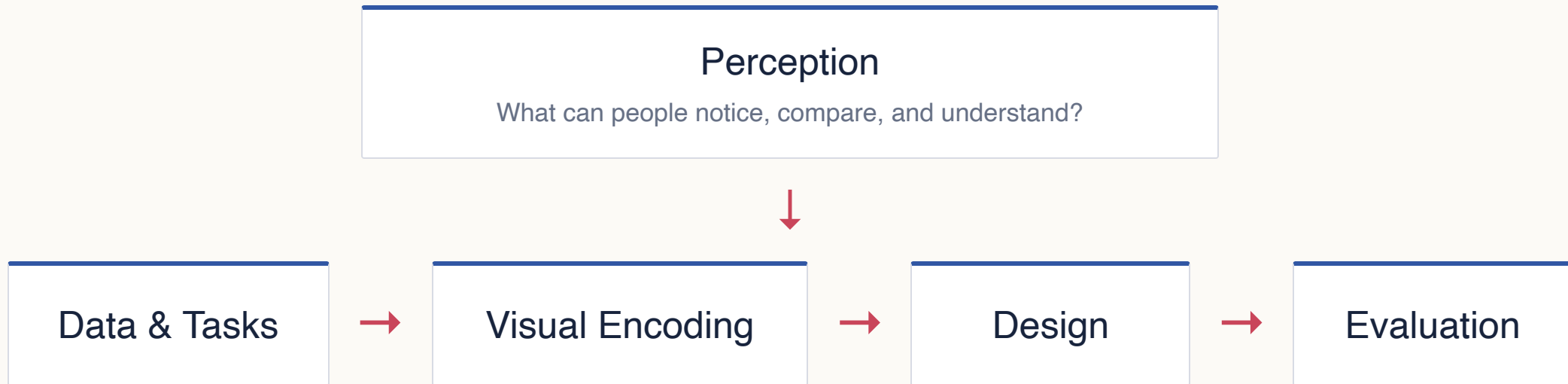
Lecture 1 — Introduction to Visualization

**Xianbin Gu**

## THE COURSE QUESTION

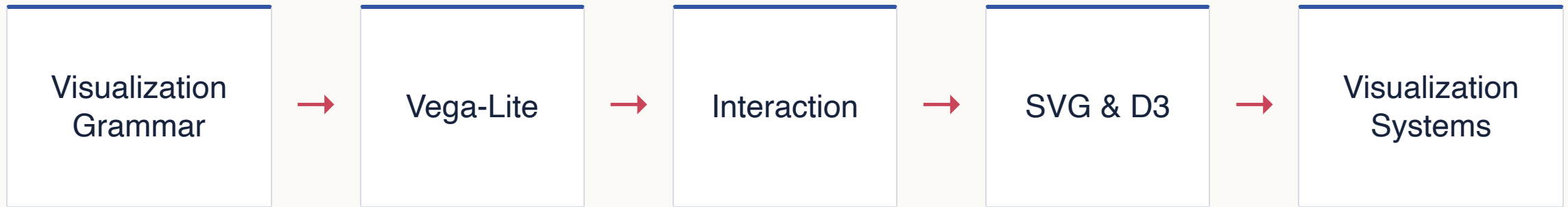
**How can we turn **data** into visual representations that help people **understand?****

# The course aims to build a visual way of thinking



A **visual way of thinking** starts with what people need to understand—not with a chart type. We then choose a visual representation that makes the important patterns easy to see.

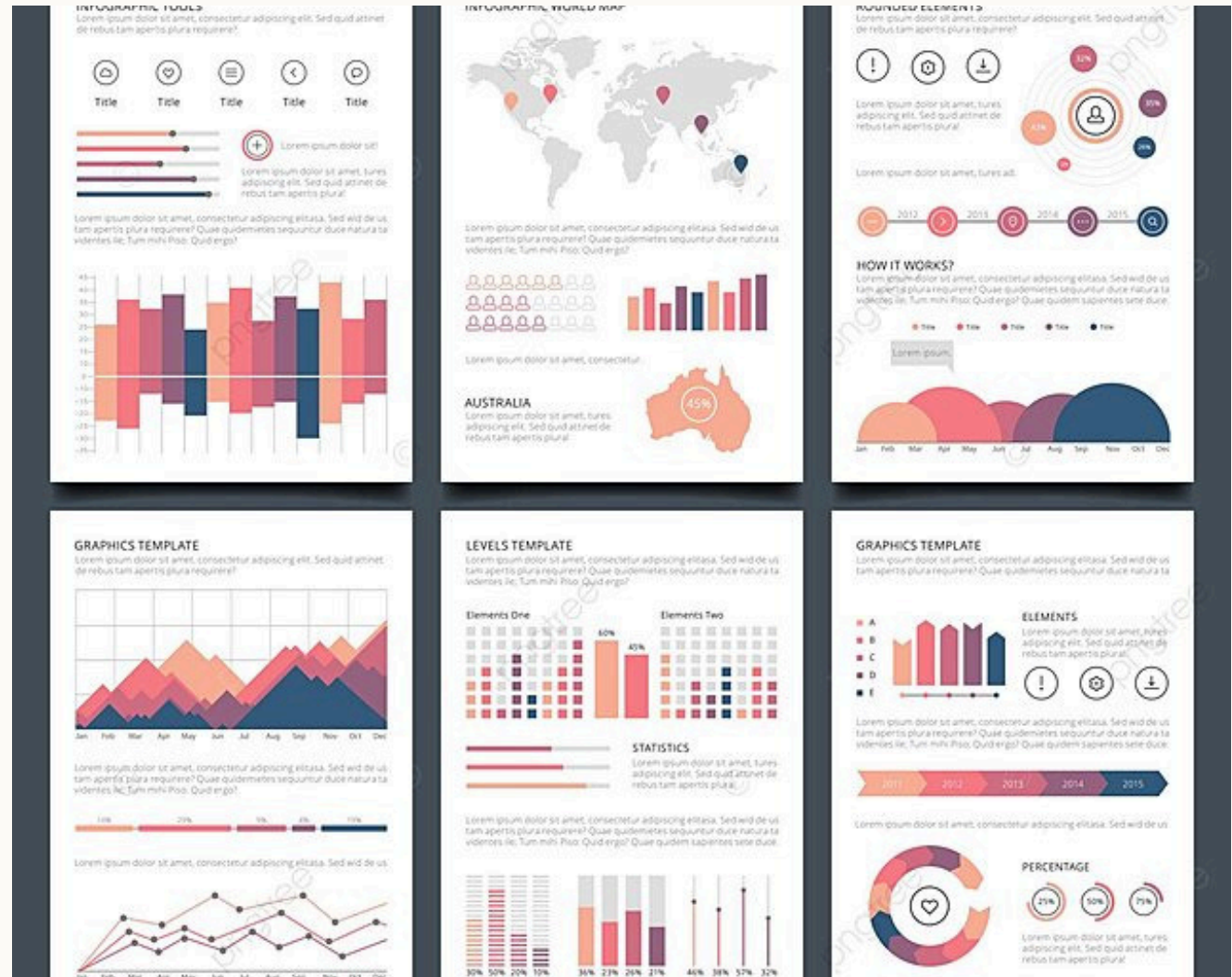
# The course also teaches you to build visualizations



When building visualizations, you will use **AI-assisted workflows** to prototype, refine, and debug—while remaining responsible for the data, design choices, and final result.

# What will you create?

- Statistical figures
- Explanatory visualization
- Interaction on figures
- Final project



# Course logistics

14

weeks

6

assignments

2

quizzes

1

students / project team

Please refer to the **syllabus on Brightspace** for more details.

# Today's agenda

- 
- |    |                               |                                                                                 |
|----|-------------------------------|---------------------------------------------------------------------------------|
| 01 | <b>What is visualization?</b> | How visual representations support exploration, explanation, and communication. |
| 02 | <b>Why do we need it?</b>     | How perception reveals patterns and reduces cognitive effort.                   |
-

## 01 · WHAT IS INFORMATION VISUALIZATION?

# What can visualization help us do?



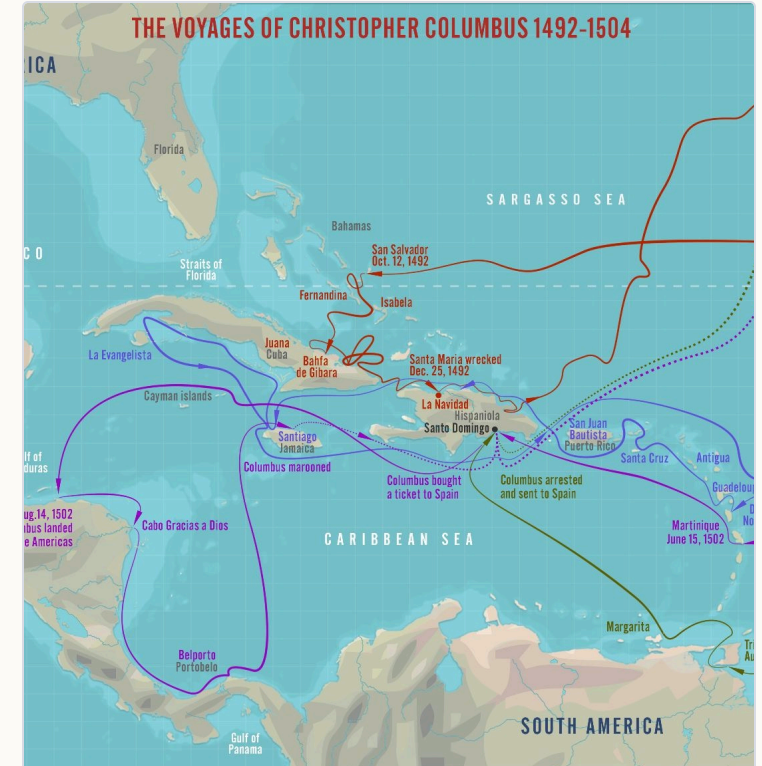
# Visualization is not new



marks that preserve experience



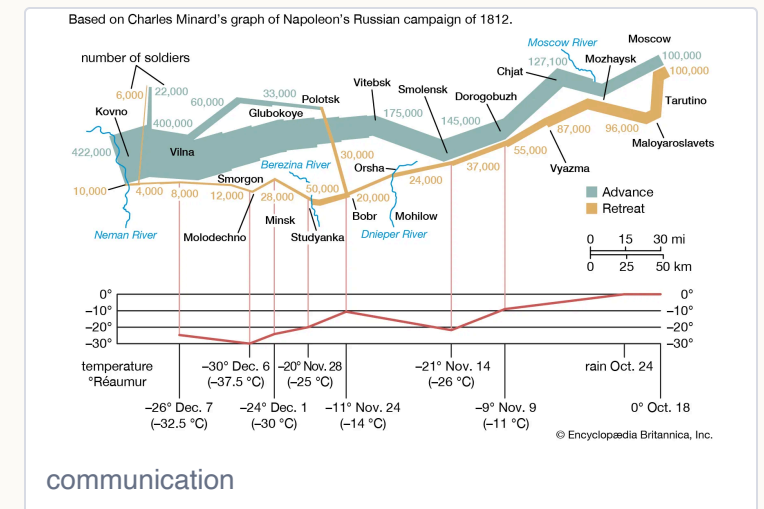
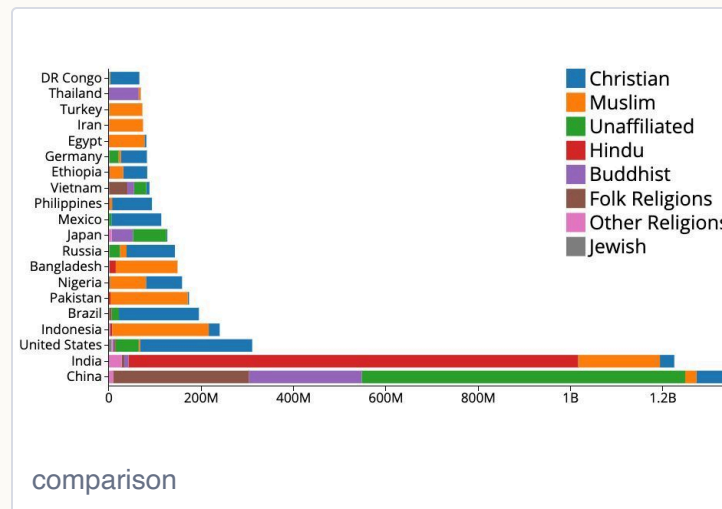
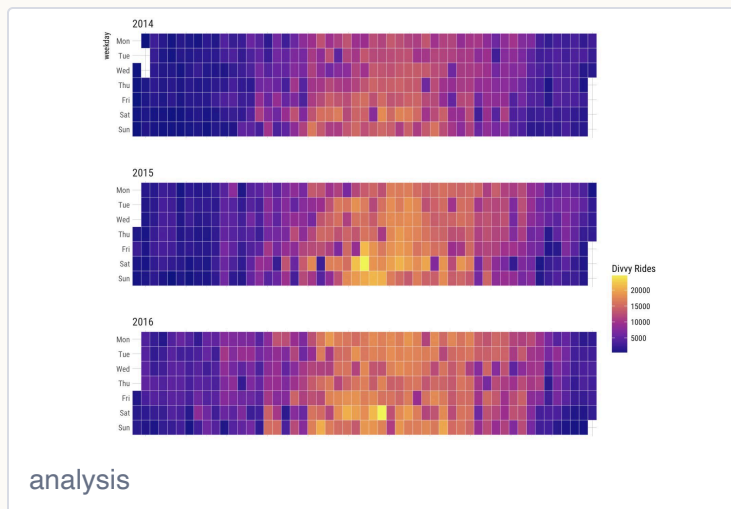
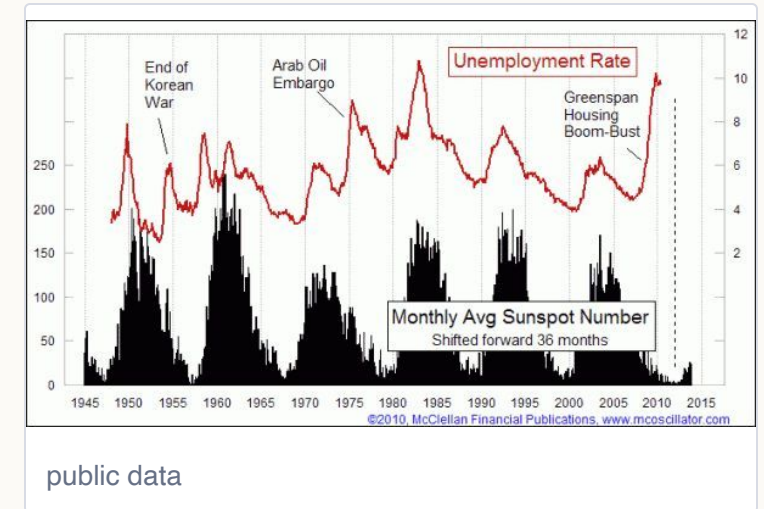
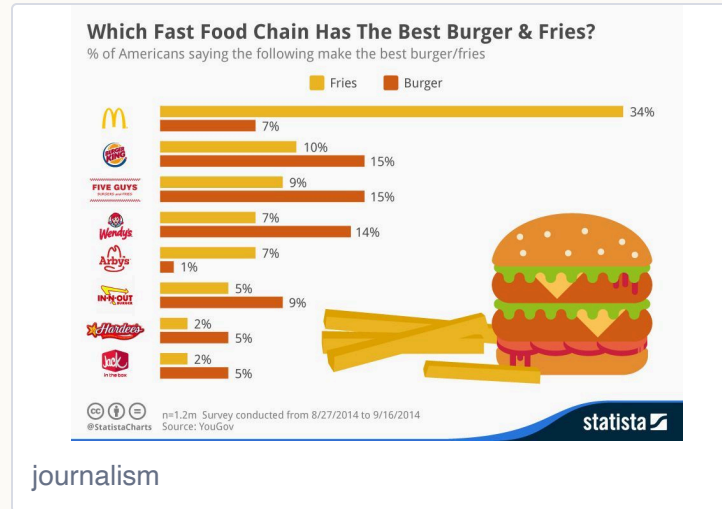
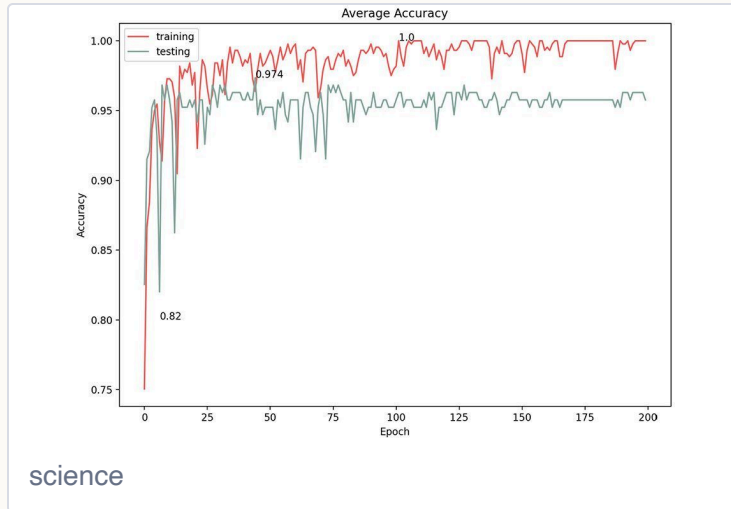
maps that organize space



maps that support navigation

**Same human impulse:** Make what we know visible—so it can be remembered, shared, and used.

# Visualization is everywhere nowadays



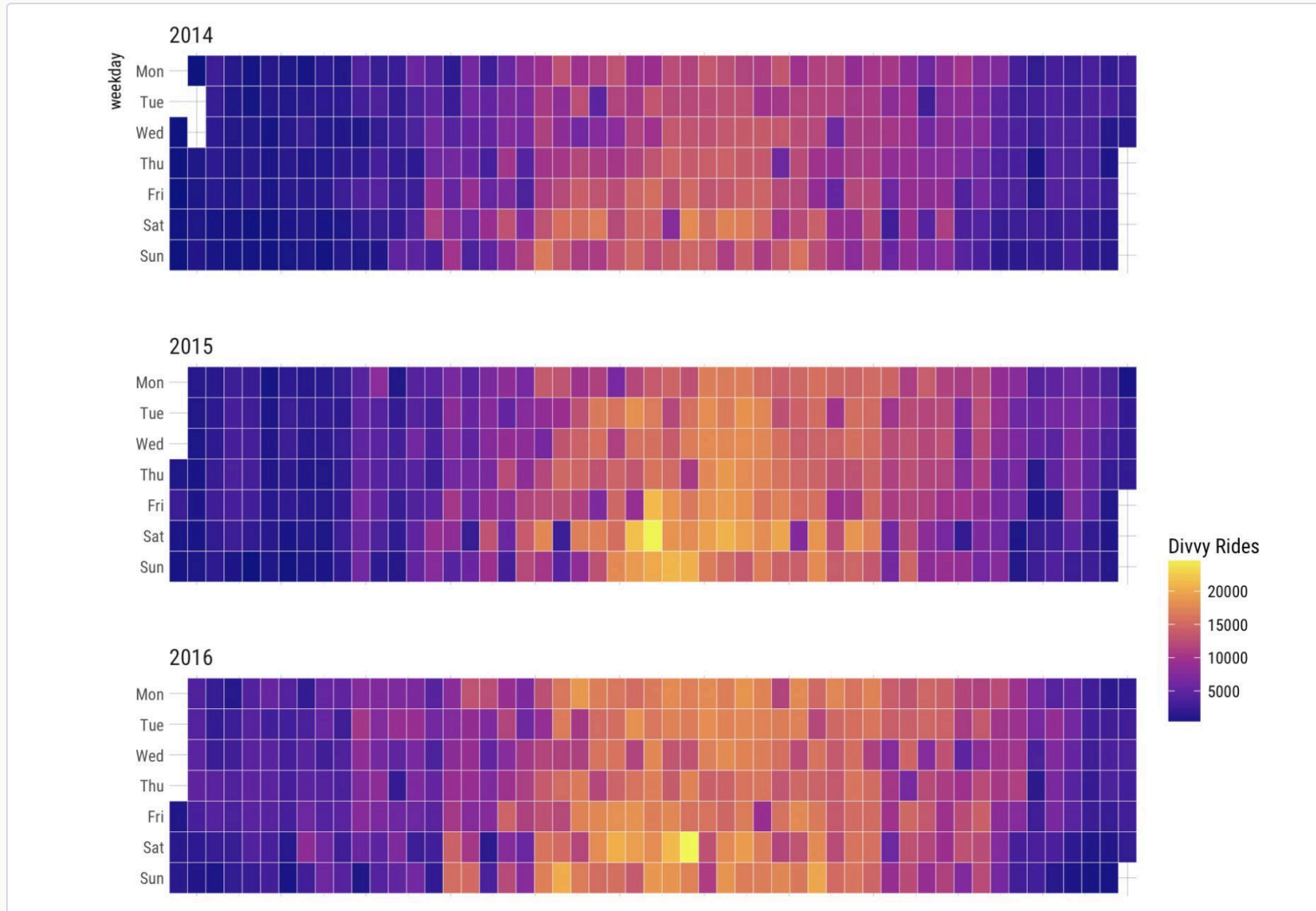
## DISCUSS

# What are these visualizations trying to accomplish?

Look for the job each visual is doing—not just its chart type.



# Explore

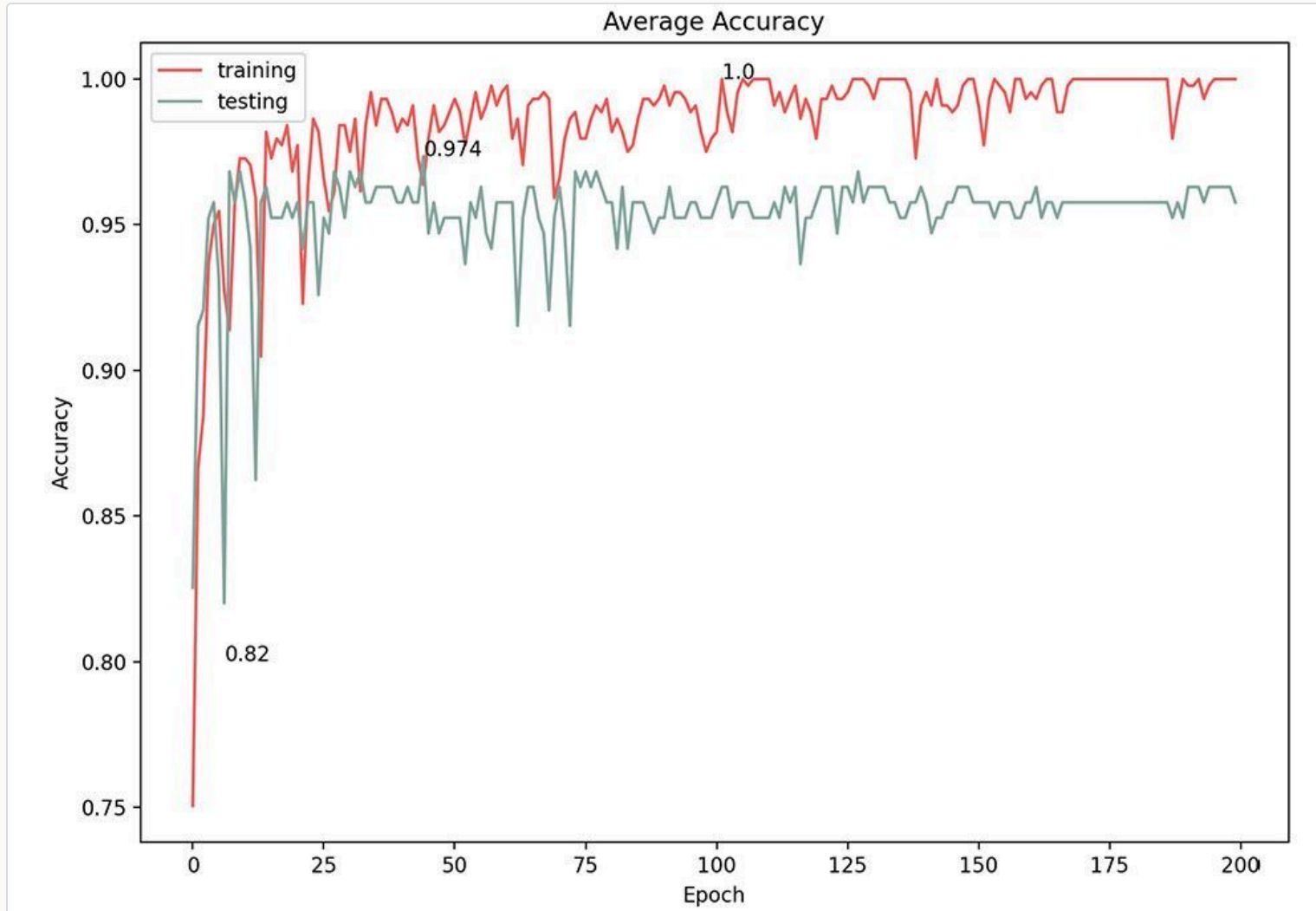


**Discover patterns we did not know to look for.**

Where are the peaks? What changed between years?

Which day behaves differently?

# Explain

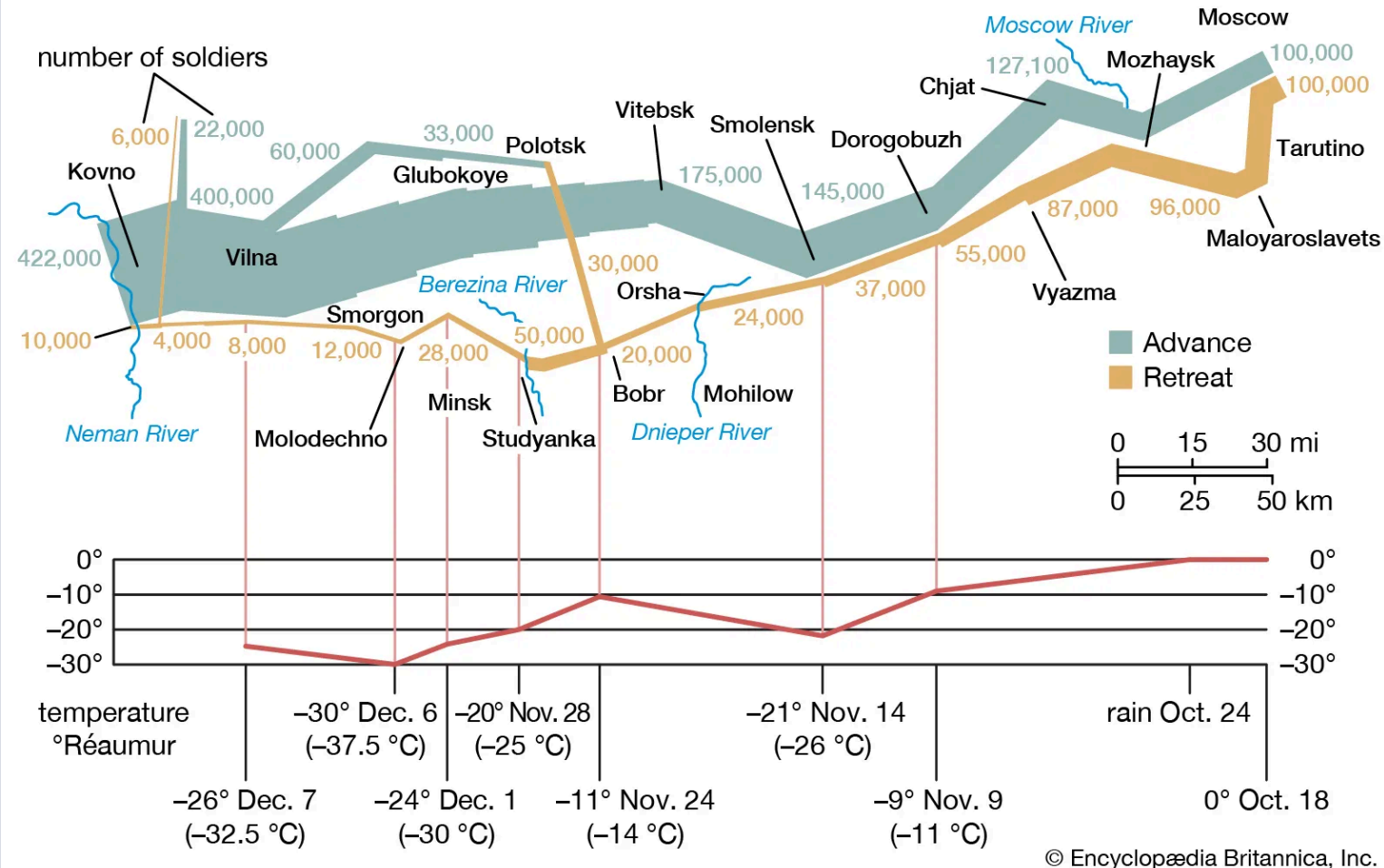


## Explain patterns and relationships.

A figure turns experimental results into evidence another person can inspect.

# Communicate

Based on Charles Minard's graph of Napoleon's Russian campaign of 1812.



**Communicate an idea or tell a story.**

One image connects geography, time, army size, direction, and temperature.

Charles Joseph Minard, Napoleon's 1812 campaign, 1869.

## A FUNCTIONAL DEFINITION OF INFORMATION VISUALIZATION

Computer-based visualization systems provide **visual representations of datasets** designed to help people carry out **tasks** more efficiently.

**DATA → VISUAL REPRESENTATION → TASK**

Tamara Munzner, *Visualization Analysis and Design*, 2014.

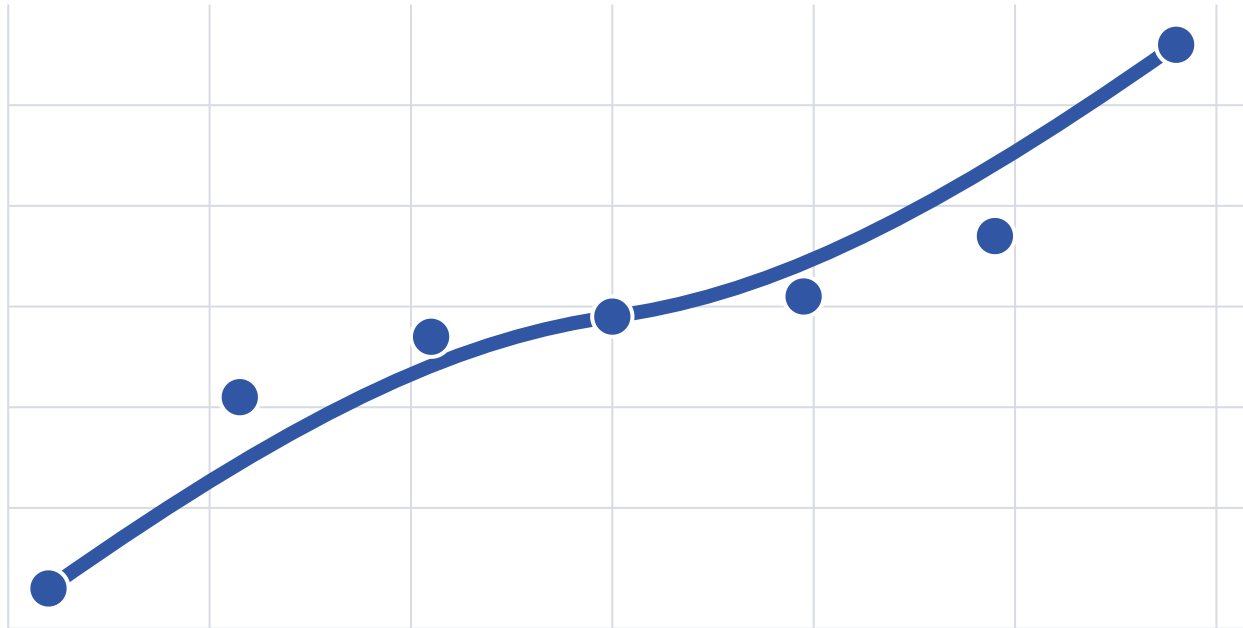
# Can you see the pattern?

|          |     |    |    |   |   |   |    |
|----------|-----|----|----|---|---|---|----|
| <b>x</b> | -3  | -2 | -1 | 0 | 1 | 2 | 3  |
| <b>y</b> | -27 | -8 | -1 | 0 | 1 | 8 | 27 |

From the table alone, you must compare values and mentally test a rule. What relationship connects x and y?



# What about now?



**Same data. Less mental work.**

The plot externalizes the comparison: the increasing curve becomes visible without checking each pair or holding the values in memory.

$$y = x^3$$

# Are these four datasets the same?

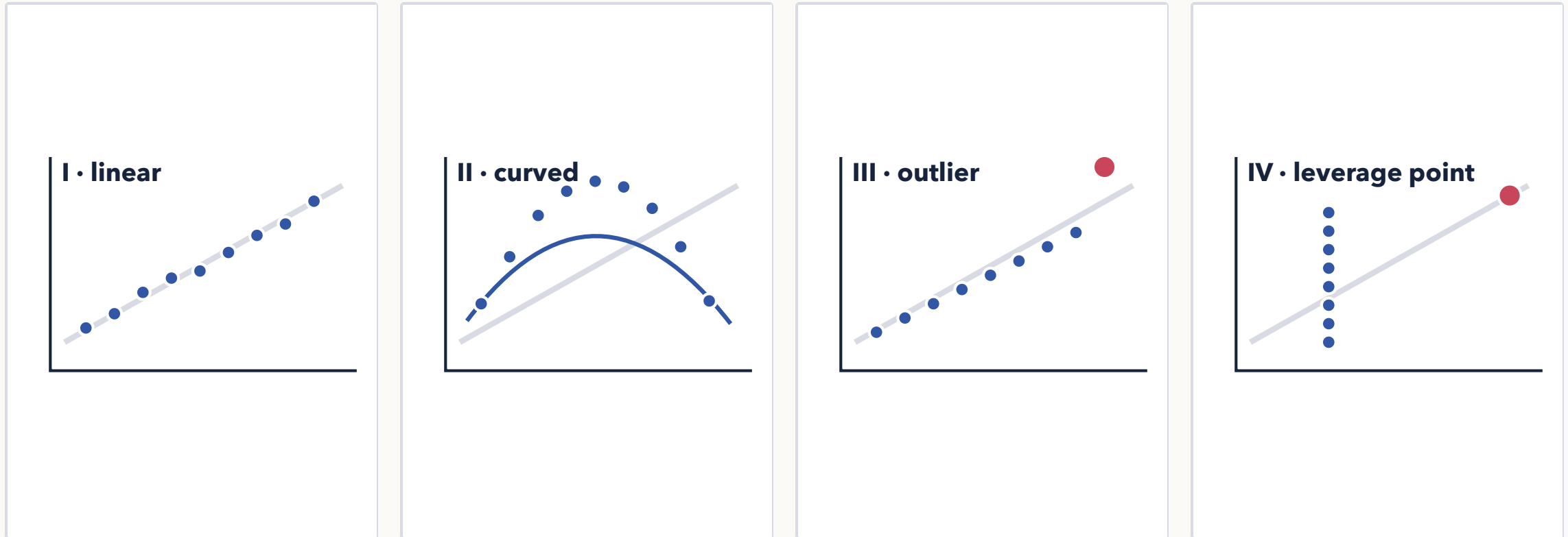
| dataset | $\bar{x}$ | $\bar{y}$ | variance x | variance y | correlation |
|---------|-----------|-----------|------------|------------|-------------|
| I       | 9.0       | 7.50      | 11.0       | 4.13       | .816        |
| II      | 9.0       | 7.50      | 11.0       | 4.13       | .816        |
| III     | 9.0       | 7.50      | 11.0       | 4.12       | .816        |
| IV      | 9.0       | 7.50      | 11.0       | 4.12       | .817        |

The summaries are nearly identical—but summaries alone cannot tell us what structure produced them.

# Same statistics...

The statistics suggest the same story. Can you identify the shape, curvature, or outliers from these numbers alone?

## ...different structures (Anscombe's quartet)

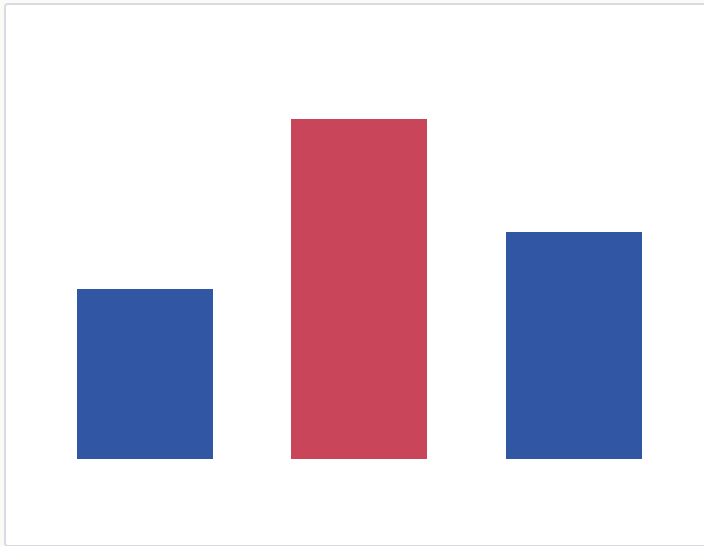


**Statistics compress—and may hide.**

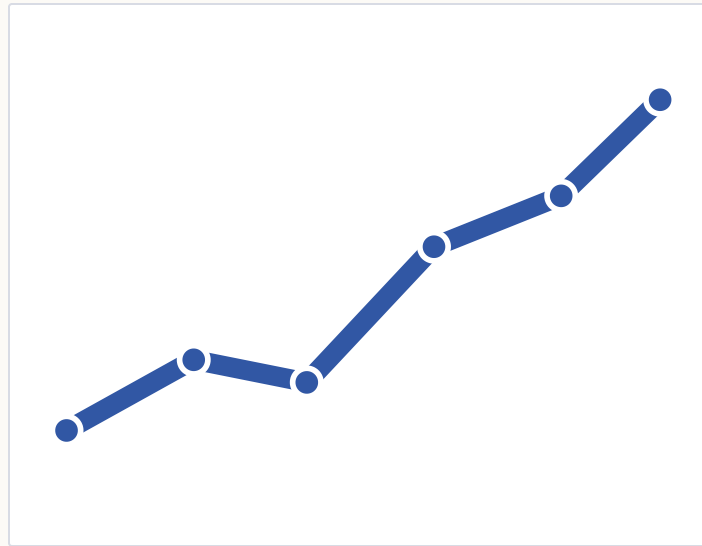
Anscombe, F. J. (1973), "Graphs in Statistical Analysis." Values shown are rounded.

**Visualization exposes structure at a glance.**

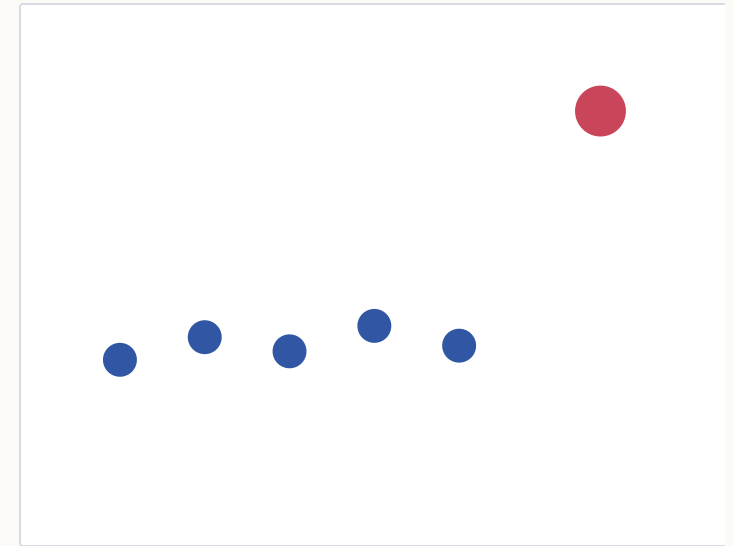
# Visualization answers questions



Which group is largest?



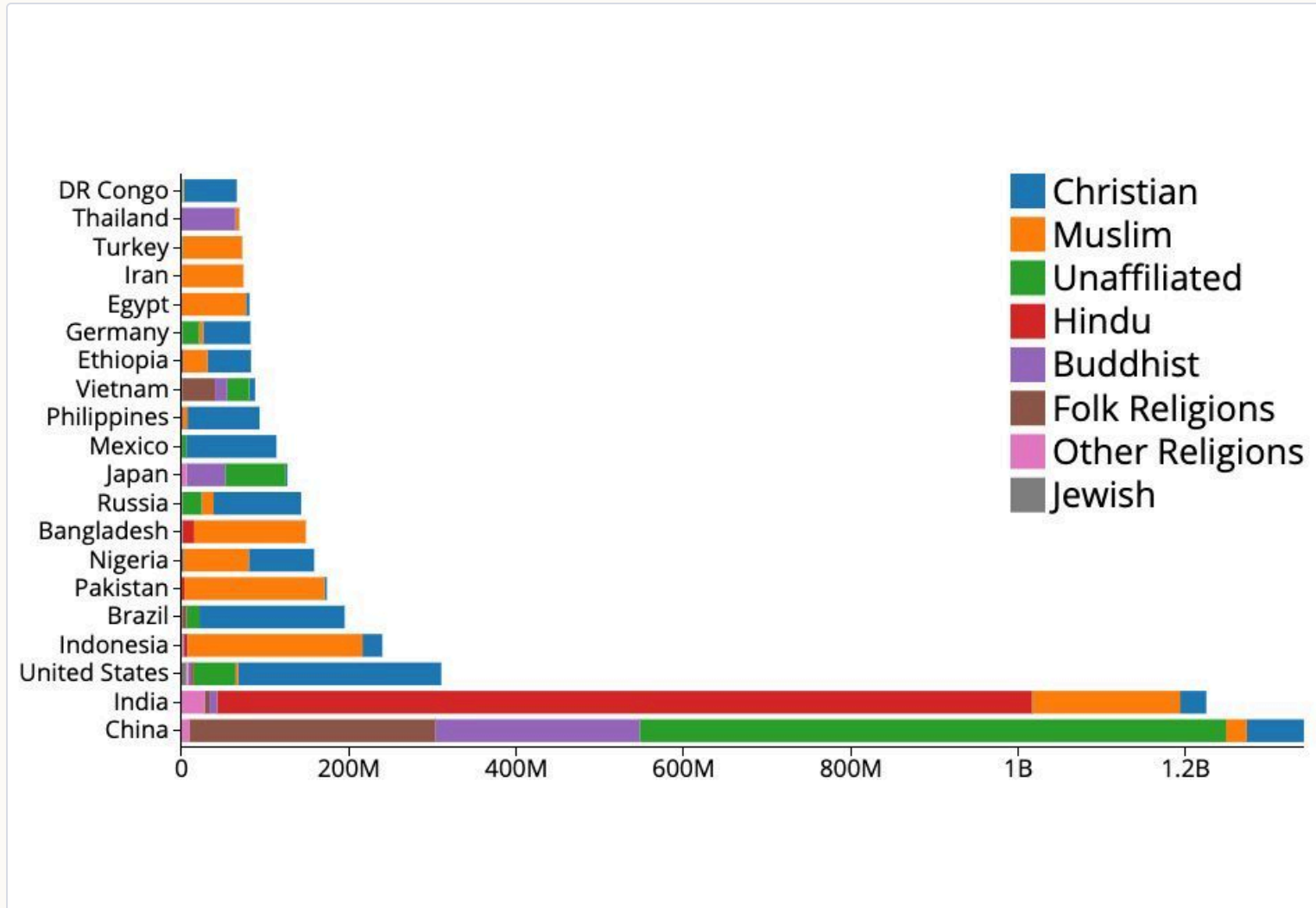
Where is the trend?



Which observation is unusual?

Position and shape turn calculation-heavy questions into rapid visual comparisons.

# Visualization generates questions efficiently



## One overview, many questions

Notice dominance and diversity, then ask why some countries are mixed while others are not.

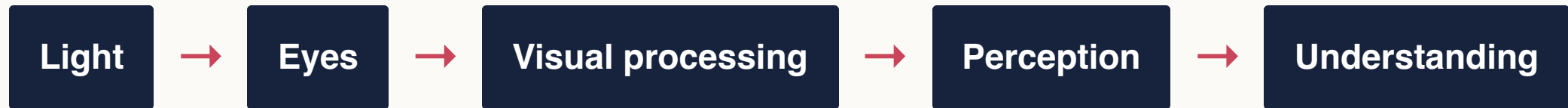
[https://hogwild.github.io/infovis2022spring/religion\\_by\\_country.html](https://hogwild.github.io/infovis2022spring/religion_by_country.html)

02 · WHY DO WE NEED VISUALIZATION?

# Why are our eyes so good at finding patterns?



# Seeing can accelerate understanding



- *Visual perception* selects, organizes, and interprets the visual signals received by the eyes.
- It turns differences in position, color, shape, and motion into patterns that people can recognize and understand.

Visualization works by shaping **visual signals** so important information is easier to perceive.

10-SECOND CHALLENGE

# Which row is larger?

47 · 83 · 16 · 92 · 58 · 31

51 · 79 · 24 · 88 · 63 · 29

Remember → compare → calculate

# Visualization externalizes information

## WITHOUT A VISUAL

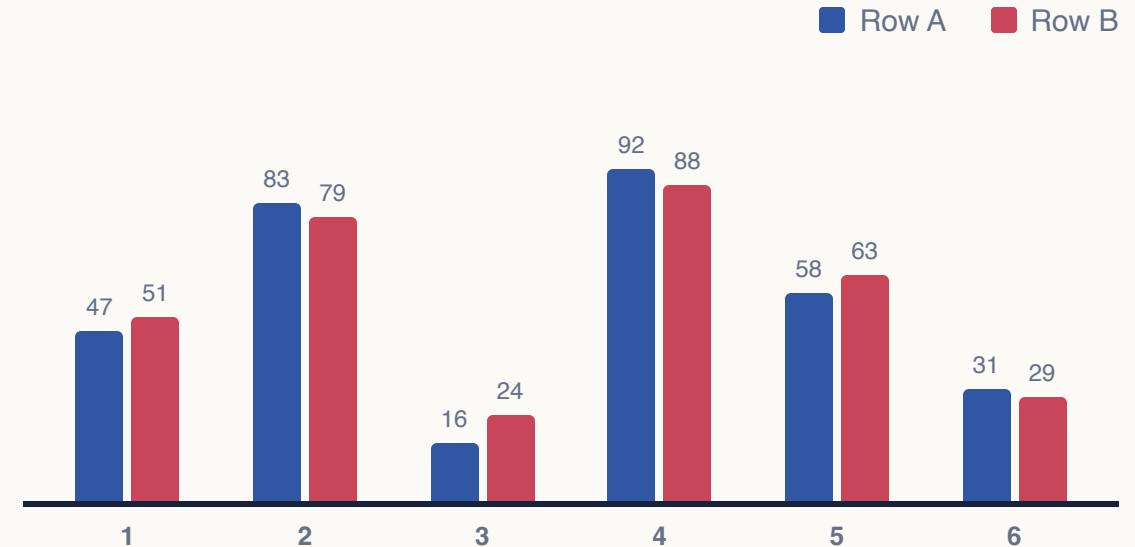
**Remember → Compare → Calculate**

Row A    47   83   16   92   58   31

Row B    51   79   24   88   63   29

## WITH A VISUAL

**Look → Compare**



**Visualization offloads mental work from memory and calculation to perception.**

# Vision is a high-bandwidth channel into human reasoning



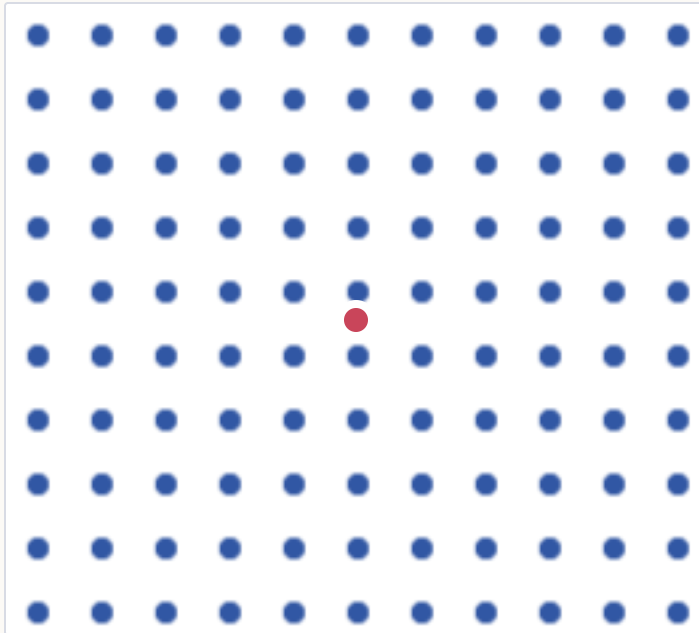
Fewer green apples



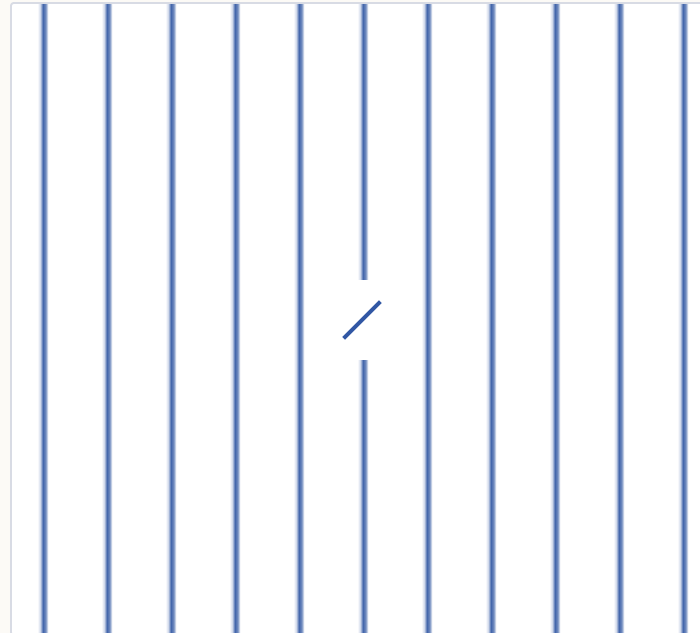
Many more green apples

**Visual perception can detect many patterns rapidly and in parallel.** The red apple pops out in both, so search time is largely independent of the number of green apples.

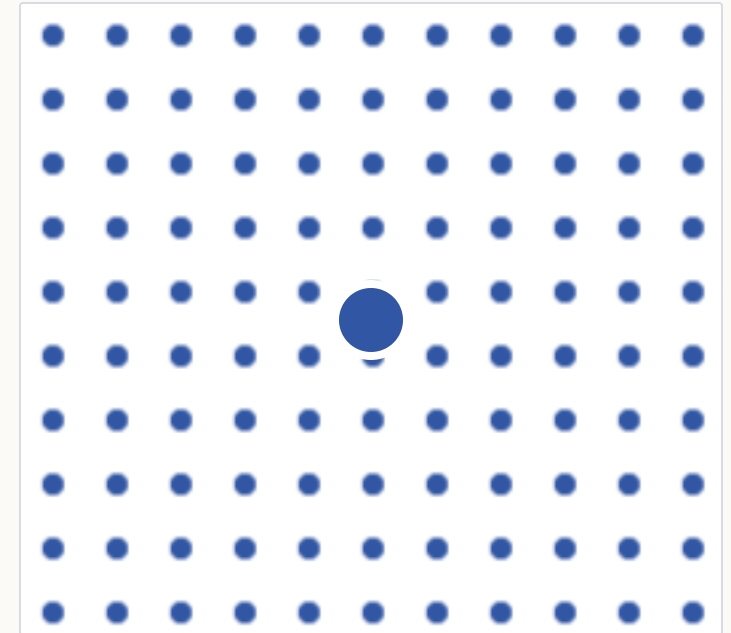
# Some visual features “pop out”



color



orientation



size

**Preattentive processing** happens rapidly—before deliberate search.

# Visualization is designed, not decorated.

- Visual attributes such as **color, position, size, and shape** can carry information.
- Different tasks often require different **visual encodings**.

# How many Vs?

**MTHIVLWYADCEQGHKILKMTWYN  
ARDCAIREQGHLVKMFPSTWYARN  
GFPSVCEILQGKMFP SNDRCEQDIFP  
SGHLMFHKMVPSTWYACEQTRN**

## How many Vs?

MTHI**V**LWYADCEQGHKILKMTWYN  
ARDCAIREQGH**L****V**KMFPSTWYARN  
GFPS**V**CEILQGKMFPSENDRCEQDIFP  
SGHLMFHKM**V**PSTWYACEQWRN

4



# Which number appears most often?

15 19 60 33 11 75 57 34 79 18 51 92  
73 22 13 71 60 22 17 10 68 73 18 55  
65 46 29 60 73 22 46 92 97 10 58 46  
57 17 83 26 99 33 88 92 60 91 29 57  
96 12 47

# Now?

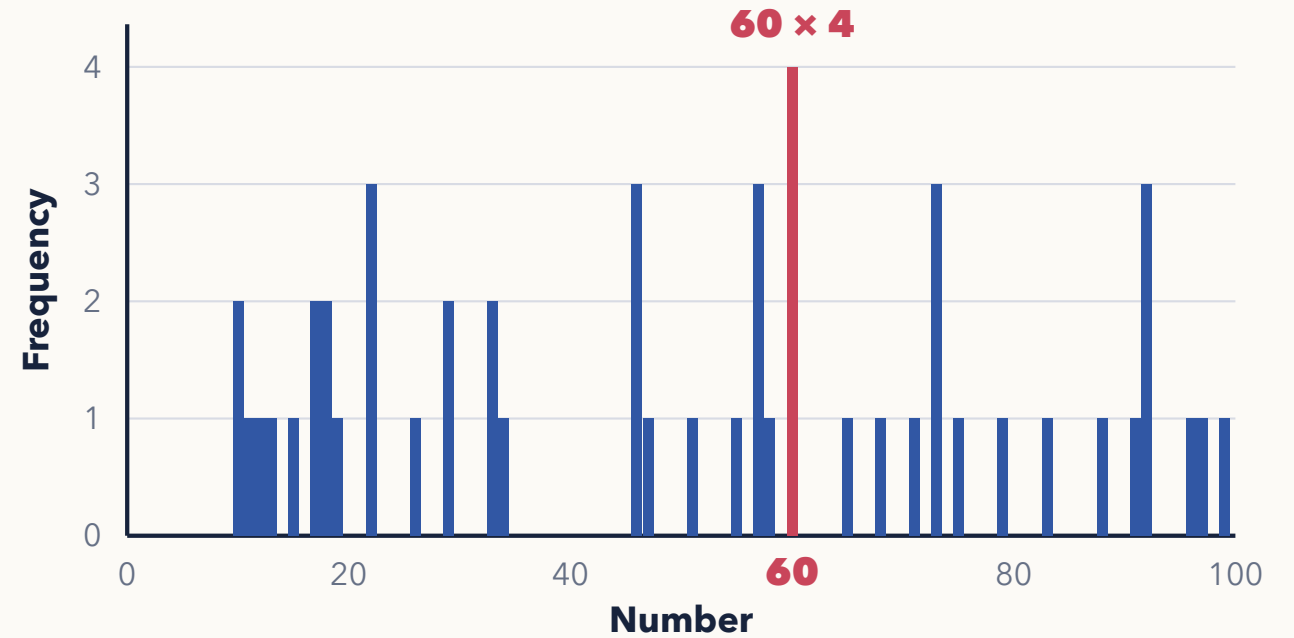
|    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|
| 15 | 19 | 60 | 33 | 11 | 75 | 57 | 34 | 79 | 18 | 51 | 92 |
| 73 | 22 | 13 | 71 | 60 | 22 | 17 | 10 | 68 | 73 | 18 | 55 |
| 65 | 46 | 29 | 60 | 73 | 22 | 46 | 92 | 97 | 10 | 58 | 46 |
| 57 | 17 | 83 | 26 | 99 | 33 | 88 | 92 | 60 | 91 | 29 | 57 |

96 12 47

60 × 4

# How about this one?

15 19 **60** 33 11 75  
57 34 79 18 51 92  
73 22 13 71 **60** 22  
17 10 68 73 18 55  
65 46 29 **60** 73 22  
46 92 97 10 58 46  
57 17 83 26 99 33  
88 92 **60** 91 29 57  
96 12 47



# Why does visualization help?

- 1 It externalizes information.**  
It shifts work from memory and calculation to visual comparison.
- 2 It reveals patterns quickly.**  
Visual features can be processed rapidly and in parallel.
- 3 It directs attention.**  
Effective encoding makes differences, patterns, and anomalies easier to find.

**Visualization helps people see what would otherwise require more mental effort.**

**Do we always need visualization?**

**No.**

# Visualization keeps humans in the loop



## Explore unfamiliar data

A researcher does not yet know which patterns or questions matter.

## Monitor and intervene

An operator must notice an anomaly and decide whether to act.

## Explain evidence

An analyst must show others why a conclusion or recommendation is justified.

Use visualization when people must participate in the reasoning process.

# When humans are not in the loop

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## High-frequency trading

Decisions occur faster than a person can inspect a chart.



## Autonomous industrial control

A trusted system senses and acts without human interpretation.

If automation can perform the task completely—and is trusted—visualization may not be necessary.

# What did we learn today?

- 1 Visualization maps **data** into visual representations.
- 2 It exploits **perception** to reduce cognitive effort.
- 3 Encoding determines what people can **see**.



**If visualization works through perception...**

**How do we see, organize, and interpret visual information?**

NEXT LECTURE

# Visual Perception & Cognition

From sensory input to visual understanding—and from perceptual principles to effective encoding.